

performance-per-watt – enhancing how much computational work you can get done for each watt of electricity used.

“Central to this effort is AMD’s ambitious 30×25 initiative, which targets a thirtyfold increase in energy efficiency for its server processors and accelerators by 2025,” Jagadish explained. According to her, “AMD-powered servers consume up to 20% less power than the competition,”



which translates into cost savings and, ideally, a smaller carbon footprint. For data centers in India, that can be a game-changer, reducing the load on local grids and enabling bigger expansions without compounding energy issues.

Bringing affordable AI access to Indian consumers

Now, all that HPC stuff is well and good, but how does it help everyday Indians who might not have top-tier internet speeds or unlimited budgets? Jaya Jagadish insisted that AMD is committed to bridging that gap. “AI is reshaping industries across the globe, and AMD is committed to playing a key role in making AI-driven services more accessible and affordable for consumers,” she said. Part of this is a nod toward India’s national strategy: “AI for All,” “Make in India,” etc. The synergy there involves AMD delivering the robust hardware that local OEMs or government agencies can leverage for advanced AI solutions.

She cited a real-world example in healthcare, specifically the E-Sanjeevani telemedicine initiative. “In the health-care sector, AMD has partnered with the Indian government to enhance the E-Sanjeevani telemedicine initiative. The virtual ophthalmology physician system, powered by our latest Turin Gen 5 EPYC CPUs, facilitates the delivery of remote healthcare services to more than

600,000 villages across India,” she said. With advanced computing, doctors can diagnose eye diseases without physically being in the same town – a huge leap for rural healthcare.

Beyond healthcare, AMD’s hardware also powers solutions in transportation (like the KAVACH 4.0 train collision avoidance system) and even the Indian space program. “AMD technology played a pivotal role in the Indian Space Research Organisation’s (ISRO) Chandrayaan-3 mission, and AMD-powered devices contributed to critical subsystems within the Vikram lander, which supported India’s historic lunar landing,” according to Jagadish. That’s no trivial bragging rights!

Because AMD also makes GPUs for gamers, I wanted to know how HPC synergy trickles down to e-sports or AAA titles. Jaya explained, “AMD is revolutionizing the online gaming experience with advanced data center technologies to deliver seamless, high-performance gameplay. AMD EPYC processors and Instinct accelerators are at the core of this innovation, which power high-performance servers capable of handling demanding workloads, including graphically intensive games.”



This means even smaller-scale dedicated game servers can handle more concurrency with minimal lag. “Gamers enjoy smooth, responsive gameplay with significantly reduced latency,” Jaya stressed, highlighting how the synergy between AI and HPC helps dynamically optimise performance, maybe even tweak rendering or manage intelligent NPC behaviours in real-time.

Indian semiconductor manufacturing aspirations

With the Indian government having approved the Semicon India pro-

gramme with a total outlay of ₹76,000 crore for the development of semiconductor and display manufacturing ecosystem in the country, India’s aspirations are big without a doubt. As a member of the Indian government’s Semicon Talent Building Initiative, I had to ask Jaya Jagadish if India would ever become a serious competitor in semiconductor manufacturing? Jaya responded, “India’s semiconductor industry is growing steadily, driven by strong government investments and a vision for a self-reliant ecosystem.”



She points out that India started this journey only about 30 years ago, compared to the 70-year head start some other countries have. But the gap is closing quickly, backed by policies encouraging local design firms and start-ups. “The next step should be to focus on nurturing and growing local design companies. Supporting startups will be crucial in this regard,” she noted. AMD itself invests in that growth through collaborations like the Society for Innovation and Entrepreneurship (SINE) at IIT Bombay, funding energy-efficient chip startups. This approach fosters the creation of homegrown IP rather than just importing everything. “These efforts solidify India’s position as a vital hub in the global semiconductor industry,” Jaya said, referencing how such development isn’t purely about assembly lines – it’s also about design expertise, local R&D labs, and honing specialised skill sets. **■**

